

LCM-Sidon Sets: A Main-Term Resolution and a Second-Order Follow-Up

Standalone write-up based on Problem 3 from the AI–Erdős problem list

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Abstract

For $N \geq 1$, write $[N] = \{1, 2, \dots, N\}$. A set $A \subseteq [N]$ is called *LCM-Sidon* if the least common multiples of unordered pairs of distinct elements of A are all distinct. The original problem asks for the maximum possible size $L(N)$ of such a set, and in particular asks whether $L(N)$ can have positive density. We prove the main-term asymptotic

$$L(N) = \pi(N) + O\left(N \exp(-c\sqrt{\log N})\right)$$

for an absolute constant $c > 0$; the proof below gives, for example, $c = 1/7$ after adjusting the implicit constant and treating bounded N separately. Hence

$$L(N) = (1 + o(1)) \frac{N}{\log N},$$

so LCM-Sidon sets have zero density. The final sections explain why the natural next problem is a second-order question of order roughly $N^{3/4}/(\log N)^{3/2}$, connected to C_3, C_4 -free graphs on primes.

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1 The original AI–Erdős problem list

The source note contains ten AI-generated Erdős-style problems in number theory and combinatorics. For context, this section records the problem statements in condensed form. The rest of the note focuses on the LCM-Sidon problem.

Problem 1. Coprime Schur-free sets. A set $A \subseteq [N]$ is coprime Schur-free if there are no $x, y, z \in A$ with

$$x + y = z, \quad (x, y) = 1.$$

Let $C(N)$ be the maximum possible size of such a set. Is

$$C(N) = \left(\frac{2}{3} + o(1) \right) N?$$

Are near-extremal examples close to unions of two prime-divisibility classes, such as $2 \mid n$ or $3 \mid n$?

Problem 2. GCD-Sidon sets. A set $A \subseteq [N]$ is GCD-Sidon if all values

$$(a, b), \quad a < b, \quad a, b \in A,$$

are distinct. Let $G(N)$ be the maximum possible size. Is

$$G(N) = N^{1/2+o(1)}?$$

Does $G(N)/\sqrt{N}$ have a limit?

Problem 3. LCM-Sidon sets. A set $A \subseteq [N]$ is LCM-Sidon if repeated pairwise lcms force the same unordered pair. Let $L(N)$ be the maximum size. Determine $L(N)$; in particular, can $L(N)$ have positive density?

Problem 4. Non-squarefree pair-sum sets. A set $A \subseteq [N]$ is non-squarefree-sum if, for every distinct $a, b \in A$, the sum $a + b$ is not squarefree. Let $Q(N)$ be the maximum possible size. Is

$$Q(N) = \left(\frac{1}{4} + o(1) \right) N?$$

Problem 5. Primitive van der Waerden numbers. A k -term arithmetic progression

$$a, a + d, \dots, a + (k - 1)d$$

is primitive if $(a, d) = 1$. Let $W^*(r, k)$ be the least N , if it exists, such that every r -coloring of $[N]$ contains a monochromatic primitive k -term arithmetic progression.

Problem 6. Prime-intersection families. A family $\mathcal{F} \subseteq 2^{[N]}$ is prime-intersecting if, for every distinct $A, B \in \mathcal{F}$, the number $|A \cap B|$ is prime. Let $P(N)$ be the maximum possible size. Is $P(N)$ polynomial in N , or can it be exponential?

Problem 7. Non-squarefree Hamming-distance codes. For $\mathcal{F} \subseteq 2^{[N]}$, define $d(A, B) = |A \Delta B|$. A family $\mathcal{F}$ is non-squarefree-distance if $d(A, B)$ is not squarefree for every distinct $A, B \in \mathcal{F}$. Let $H(N)$ be the maximum possible size. Is

$$H(N) = 2^{N/2+o(N)}?$$

Problem 8. Coprime representation of graphs. For a finite graph G , let $\rho(G)$ be the least M such that the vertices of G can be injectively labelled by integers in $[M]$ with

$$uv \in E(G) \iff (\ell(u), \ell(v)) = 1.$$

Let $R(n) = \max_{|V(G)|=n} \rho(G)$. Is

$$R(n) = 2^{\Theta(n)}?$$

Problem 9. The squarefree divisor graph. Let D_N be the graph on $[N]$ in which $a < b$ are adjacent if $a \mid b$ and b/a is squarefree. Let $\chi(N) = \chi(D_N)$. Is

$$\chi(N) = \Theta\left(\frac{\log N}{\log \log N}\right)?$$

Problem 10. Prime-multiple discrepancy. Choose signs $\varepsilon_n \in \{-1, +1\}$ for $1 \leq n \leq N$. Define

$$D(N) = \min_{\varepsilon_1, \dots, \varepsilon_N} \max_{\substack{p \leq N \\ p \text{ prime}}} \left| \sum_{\substack{n \leq N \\ p \mid n}} \varepsilon_n \right|.$$

Is $D(N)$ bounded?

2 Problem 3 and the main theorem

We now isolate Problem 3.

Problem 1 (Original Problem 3: LCM-Sidon sets). *Call $A \subseteq [N]$ LCM-Sidon if*

$$\text{lcm}(a, b) = \text{lcm}(c, d), \quad a < b, \quad c < d, \quad a, b, c, d \in A,$$

implies

$$\{a, b\} = \{c, d\}.$$

Equivalently, the map

$$\{a, b\} \mapsto \text{lcm}(a, b)$$

is injective on the unordered two-element subsets of A . Let

$$L(N) = \max\{|A| : A \subseteq [N] \text{ is LCM-Sidon}\}.$$

Determine $L(N)$. In particular, can $L(N)$ have positive density?

The answer to the positive-density question is no. More precisely, the primes give the correct main term.

Theorem 2 (Main-term resolution). *There are absolute constants $C > 0$ and N_0 such that, for every $N \geq N_0$,*

$$\pi(N) \leq L(N) \leq \pi(N) + CN \exp\left(-\frac{1}{7}\sqrt{\log N}\right).$$

Consequently

$$L(N) = \pi(N) + O\left(N \exp(-c\sqrt{\log N})\right)$$

for an absolute constant $c > 0$, and hence

$$L(N) = (1 + o(1)) \frac{N}{\log N}.$$

In particular, LCM-Sidon subsets of $[N]$ cannot have positive density.

Remark 3. The value $1/7$ is not intrinsic. The argument naturally gives an exponent $1/6$ before absorbing a factor $O(\log \log N)$. Replacing $1/7$ by any fixed $c < 1/6$ is possible for all sufficiently large N . The exponential error is not expected to be the true second-order term.

3 The lower bound: primes are LCM-Sidon

Proposition 4 (Elementary prime construction). *For every N ,*

$$L(N) \geq \pi(N).$$

Proof. Let A be the set of all primes at most N . For distinct primes $p < q$,

$$\text{lcm}(p, q) = pq.$$

If $p < q$ and $r < s$ are primes and

$$\text{lcm}(p, q) = \text{lcm}(r, s),$$

then

$$pq = rs.$$

By unique factorisation, the two multisets of prime factors are equal, so

$$\{p, q\} = \{r, s\}.$$

Therefore the map $\{p, q\} \mapsto \text{lcm}(p, q)$ is injective on unordered pairs of distinct primes. Thus the primes up to N form an LCM-Sidon set of size $\pi(N)$. $\square$

4 The upper bound

Let $A \subseteq [N]$ be an arbitrary LCM-Sidon set. We prove

$$|A| \leq \pi(N) + O\left(N \exp\left(-\frac{1}{7}\sqrt{\log N}\right)\right).$$

4.1 Notation

For a positive integer n , write $P^+(n)$ for the largest prime factor of n , with the convention

$$P^+(1) = 1.$$

Put

$$X = \log N, \quad \beta = X^{-1/2}, \quad y = N^\beta = \exp(\sqrt{\log N}).$$

We divide elements of A according to their largest prime factor.

4.2 Few smooth elements

An integer n is y -smooth if $P^+(n) \leq y$.

Lemma 5 (Smooth numbers are negligible). *For all sufficiently large N ,*

$$\#\{n \leq N : P^+(n) \leq y\} \ll N \exp\left(-\frac{1}{2}\sqrt{\log N}\right).$$

Proof. Let

$$\Psi(N, y) = \#\{n \leq N : P^+(n) \leq y\}.$$

We use Rankin's trick. Set

$$\sigma = 1 - \frac{1}{\log y}.$$

Since $y = \exp(\sqrt{X})$, this is

$$\sigma = 1 - \frac{1}{\sqrt{X}}.$$

For each y -smooth $n \leq N$,

$$1 \leq \left(\frac{N}{n}\right)^\sigma.$$

Therefore, after extending the sum to all y -smooth positive integers,

$$\begin{aligned} \Psi(N, y) &\leq \sum_{\substack{n \geq 1 \\ P^+(n) \leq y}} \left(\frac{N}{n}\right)^\sigma \\ &= N^\sigma \sum_{\substack{n \geq 1 \\ P^+(n) \leq y}} n^{-\sigma} \\ &= N^\sigma \prod_{p \leq y} (1 - p^{-\sigma})^{-1}. \end{aligned}$$

The first factor is

$$N^\sigma = N \exp\left(-\frac{\log N}{\log y}\right) = N \exp(-\sqrt{X}).$$

It remains to bound the Euler product. If $p \leq y$, then

$$p^{-\sigma} = p^{-1} p^{1/\log y} \leq e p^{-1}.$$

For N large, $\sigma \geq 1/2$, so $p^{-\sigma} \leq 2^{-1/2} < 1$ for every prime p . Hence

$$-\log(1 - t) \ll t \quad (0 \leq t \leq 2^{-1/2})$$

gives

$$\begin{aligned} \log \prod_{p \leq y} (1 - p^{-\sigma})^{-1} &\ll \sum_{p \leq y} p^{-\sigma} \\ &\leq e \sum_{p \leq y} \frac{1}{p} \\ &\ll \log \log y. \end{aligned}$$

Thus

$$\prod_{p \leq y} (1 - p^{-\sigma})^{-1} \ll (\log y)^C$$

for an absolute constant C . Since $\log y = \sqrt{X}$, the factor $(\log y)^C$ is swallowed by half of the exponential saving:

$$N e^{-\sqrt{X}} (\log y)^C \ll N e^{-\sqrt{X}/2}$$

for all sufficiently large N . This proves the lemma. $\square$

Therefore the number of elements $a \in A$ with $P^+(a) \leq y$ is

$$O\left(N \exp\left(-\frac{1}{2}\sqrt{\log N}\right)\right).$$

4.3 Few elements whose largest prime factor occurs with exponent at least two

Lemma 6 (Repeated large prime factors are negligible). *The number of integers $n \leq N$ for which there exists a prime $p > y$ with $p^2 \mid n$ is*

$$O(N/y) = O\left(N \exp(-\sqrt{\log N})\right).$$

Consequently the number of $n \leq N$ such that $P^+(n) > y$ and $P^+(n)^2 \mid n$ is also

$$O\left(N \exp(-\sqrt{\log N})\right).$$

Proof. By the union bound,

$$\begin{aligned} \#\{n \leq N : \exists p > y, p^2 \mid n\} &\leq \sum_{p > y} \left\lfloor \frac{N}{p^2} \right\rfloor \\ &\leq N \sum_{m > y} \frac{1}{m^2} \\ &\ll \frac{N}{y}. \end{aligned}$$

Since $y = \exp(\sqrt{\log N})$, the asserted bound follows. $\square$

4.4 Good elements and the sets S_p

After discarding the elements counted in Lemmas 5 and 6, every remaining element $a \in A$ has a unique representation

$$a = pu$$

where

$$p = P^+(a) > y, \quad p \nmid u, \quad P^+(u) < p.$$

For each prime $p > y$, define

$$S_p = \{u \geq 1 : pu \in A, P^+(pu) = p, p^2 \nmid pu\}.$$

Thus the number of good elements of A is

$$\sum_{p > y} |S_p|.$$

The key observation is that the LCM-Sidon condition forces the sets S_p to have very small pairwise intersections.

Lemma 7 (Almost-disjoint cofactor sets). *If $p \neq q$ are primes, then*

$$|S_p \cap S_q| \leq 1.$$

Proof. Suppose, for contradiction, that there are two distinct positive integers $u \neq v$ with

$$u, v \in S_p \cap S_q.$$

Then

$$pu, \quad pv, \quad qu, \quad qv$$

all belong to A .

First, these four elements are distinct. The equalities $pu = pv$ or $qu = qv$ would force $u = v$, while $pu = qu$ would force $p = q$. If $pu = qv$, then, assuming $p < q$, the prime q divides u , contradicting $P^+(u) < p < q$ from $u \in S_p$. The case $q < p$ is symmetric.

Since $u \in S_p$, the prime p does not divide u ; since $u \in S_q$, the prime q does not divide u . The same is true for v . Therefore neither p nor q appears in u or v , and hence

$$\text{lcm}(pu, qv) = pq \text{ lcm}(u, v).$$

Similarly,

$$\text{lcm}(pv, qu) = pq \text{ lcm}(u, v).$$

Thus

$$\text{lcm}(pu, qv) = \text{lcm}(pv, qu).$$

But the unordered pairs

$$\{pu, qv\} \quad \text{and} \quad \{pv, qu\}$$

are distinct: equality of these pairs would imply either $u = v$ or $p = q$. This contradicts the LCM-Sidon property of A . Hence $|S_p \cap S_q| \leq 1$. $\square$

4.5 A finite incidence estimate

We shall repeatedly use the following elementary packing estimate.

Lemma 8 (Packing estimate). *Let $S_1, \dots, S_r$ be subsets of an M -element set. Suppose*

$$|S_i \cap S_j| \leq 1 \quad (i \neq j).$$

Then

$$\sum_{i=1}^r |S_i| \leq r + M\sqrt{r}.$$

Proof. Every two-element subset of the underlying M -element set can lie in at most one of the sets S_i . Therefore

$$\sum_{i=1}^r \binom{|S_i|}{2} \leq \binom{M}{2} \leq \frac{M^2}{2}.$$

For each i , put

$$t_i = \max(|S_i| - 1, 0).$$

If $|S_i| = 0$ or 1 , then $t_i = 0$. If $|S_i| \geq 2$, then

$$t_i^2 = (|S_i| - 1)^2 \leq |S_i|(|S_i| - 1) = 2 \binom{|S_i|}{2}.$$

Hence

$$\sum_{i=1}^r t_i^2 \leq 2 \sum_{i=1}^r \binom{|S_i|}{2} \leq M^2.$$

By Cauchy's inequality,

$$\sum_{i=1}^r t_i \leq \sqrt{r} \left(\sum_{i=1}^r t_i^2 \right)^{1/2} \leq M\sqrt{r}.$$

Finally,

$$|S_i| \leq 1 + t_i$$

for every i , and so

$$\sum_{i=1}^r |S_i| \leq r + \sum_{i=1}^r t_i \leq r + M\sqrt{r}.$$

□

4.6 Stratifying the largest prime factor

We now split the range $p > y$ into multiplicative intervals. Let

$$\alpha_0 = 1, \quad \alpha_{j+1} = \max(\beta, 2\alpha_j/3),$$

and stop when $\alpha_J = \beta$. The intervals

$$N^{\alpha_{j+1}} < p \leq N^{\alpha_j}, \quad 0 \leq j < J,$$

cover all primes p with $p > y = N^\beta$ and $p \leq N$.

There are only $O(\log \log N)$ such intervals. Indeed, before the sequence reaches β , it is multiplied by $2/3$ at each step, so

$$J \ll \log(1/\beta) \ll \log \log N.$$

Fix one interval. Write

$$\alpha = \alpha_j, \quad \gamma = \alpha_{j+1}.$$

Then

$$\gamma \geq \frac{2\alpha}{3}, \quad \alpha \geq \beta.$$

Let r be the number of primes in this interval:

$$r = \#\{p \text{ prime} : N^\gamma < p \leq N^\alpha\}.$$

For each such prime p , every $u \in S_p$ satisfies

$$u \leq \frac{N}{p} < N^{1-\gamma}.$$

Thus the sets S_p are subsets of an interval of length at most

$$M = N^{1-\gamma}.$$

By Lemma 7, these sets have pairwise intersections of size at most one. Applying Lemma 8 gives

$$\sum_{N^\gamma < p \leq N^\alpha} |S_p| \leq r + N^{1-\gamma} \sqrt{r}.$$

Using the crude prime number theorem estimate

$$\pi(x) \ll \frac{x}{\log x} \quad (x \geq 3),$$

we have

$$r \leq \pi(N^\alpha) \ll \frac{N^\alpha}{\alpha \log N}.$$

In particular,

$$\sqrt{r} \ll N^{\alpha/2}.$$

Therefore

$$N^{1-\gamma} \sqrt{r} \ll N^{1-\gamma+\alpha/2}.$$

Since $\gamma \geq 2\alpha/3$,

$$1 - \gamma + \frac{\alpha}{2} \leq 1 - \frac{2\alpha}{3} + \frac{\alpha}{2} = 1 - \frac{\alpha}{6}.$$

Since $\alpha \geq \beta = X^{-1/2}$,

$$N^{-\alpha/6} \leq N^{-\beta/6} = \exp\left(-\frac{1}{6}\sqrt{\log N}\right).$$

Thus the total contribution from this interval is at most

$$r + O\left(N \exp\left(-\frac{1}{6}\sqrt{\log N}\right)\right).$$

4.7 Completion of the proof of Theorem 2

Summing over all $J = O(\log \log N)$ intervals gives

$$\begin{aligned} \sum_{p > y} |S_p| &\leq \sum_{j=0}^{J-1} r_j + O\left((\log \log N) N \exp\left(-\frac{1}{6}\sqrt{\log N}\right)\right) \\ &\leq \pi(N) + O\left(N \exp\left(-\frac{1}{7}\sqrt{\log N}\right)\right). \end{aligned}$$

Here the r_j 's count disjoint sets of primes, so their sum is at most $\pi(N)$, and the factor $\log \log N$ is absorbed by weakening $1/6$ to $1/7$.

We now add back the discarded exceptional elements. By Lemma 5, the y -smooth elements contribute

$$O\left(N \exp\left(-\frac{1}{2}\sqrt{\log N}\right)\right).$$

By Lemma 6, the elements whose largest prime factor exceeds y but occurs with exponent at least two contribute

$$O\left(N \exp(-\sqrt{\log N})\right).$$

All remaining elements are counted by $\sum_{p > y} |S_p|$. Hence

$$\begin{aligned} |A| &\leq \pi(N) + O\left(N \exp\left(-\frac{1}{7}\sqrt{\log N}\right)\right) \\ &\quad + O\left(N \exp\left(-\frac{1}{2}\sqrt{\log N}\right)\right) + O\left(N \exp(-\sqrt{\log N})\right) \\ &= \pi(N) + O\left(N \exp\left(-\frac{1}{7}\sqrt{\log N}\right)\right). \end{aligned}$$

Since A was arbitrary, this proves the desired upper bound. Proposition 4 gives the matching lower bound $\pi(N)$, completing the proof of Theorem 2.

5 What the proof does and does not determine

The theorem resolves the main term and the positive-density question. It does not determine the exact value of $L(N)$, nor the true second-order term.

The error term

$$O\left(N \exp(-c\sqrt{\log N})\right)$$

comes from a deliberately coarse decomposition by largest prime factor. This is strong enough to show that $L(N) = o(N)$, and even that $L(N) \sim N/\log N$, but it is much larger than the natural lower-order constructions described below.

The proof suggests the following heuristic. Most elements of a large LCM-Sidon set should be primes. Any substantial excess over $\pi(N)$ should come from carefully chosen composite numbers, especially semiprimes pq , whose prime factors act like vertices of a graph. Repeated lcms among pairs of semiprimes correspond to short cycles or triangles in that graph.

6 A lower-order construction from C_3, C_4 -free graphs

The primes are not exactly optimal.

Proposition 9 (Semiprime graph construction). *Let G be a graph whose vertices are primes at most $\sqrt{N}$. Suppose G has no triangle and no 4-cycle. Define*

$$A_G = (\{p \leq N : p \text{ prime}\} \setminus V(G)) \cup \{pq : pq \in E(G)\}.$$

Then A_G is LCM-Sidon. Consequently, if $m = \pi(\sqrt{N})$, then

$$L(N) \geq \pi(N) - m + \text{ex}(m, \{C_3, C_4\}).$$

Proof. Every element of A_G is squarefree. Its prime support is either a singleton $\{r\}$, where r is a prime not used as a vertex of G , or an edge $\{p, q\}$ of G . Singleton supports and edge supports are separated because the primes in $V(G)$ have been removed from the prime part of A_G .

Suppose

$$\text{lcm}(x, y) = \text{lcm}(z, w)$$

for four elements $x, y, z, w \in A_G$, with $x \neq y$ and $z \neq w$. We need to prove

$$\{x, y\} = \{z, w\}.$$

Let $S(t)$ denote the prime support of t . Since all elements of A_G are squarefree, equality of lcms is the same as equality of unions of prime supports:

$$S(x) \cup S(y) = S(z) \cup S(w).$$

First suppose this common union contains a prime $r \notin V(G)$. Such a prime can occur only as the singleton support $\{r\}$ of the prime element $r \in A_G$. It cannot occur inside any semiprime from A_G , since semiprimes use only vertices of G . Hence the element r must appear among x, y and also among z, w . Removing this common singleton from both unordered pairs leaves equality of the other support, so the other element is also the same.

It remains to consider the case in which the common union contains only primes from $V(G)$. Then all four elements involved are semiprimes corresponding to edges of G . We use the following graph fact:

In a graph with no triangles and no 4-cycles, two unordered pairs of distinct edges cannot have the same union of endpoints unless the two edge-pairs are identical.

To prove the fact, let e_1, e_2 and f_1, f_2 be two unordered pairs of distinct edges such that

$$e_1 \cup e_2 = f_1 \cup f_2$$

as vertex sets. If e_1 and e_2 share a vertex, their union has three vertices and forms a path of length two. Any different pair of edges on the same three vertices would require the third edge joining the two endpoints of the path, creating a triangle. If e_1 and e_2 are disjoint, their union has four vertices. A different pair of edges with the same four-vertex union would be another perfect matching on these four vertices; the two matchings together form a 4-cycle. Both alternatives are forbidden, so the edge-pair is unique.

Therefore the two semiprime pairs are the same. In every case, equality of lcms forces equality of unordered pairs, so A_G is LCM-Sidon.

Its size is

$$|A_G| = \pi(N) - |V(G)| + |E(G)|.$$

Taking G to be an extremal C_3, C_4 -free graph on the $m = \pi(\sqrt{N})$ primes at most $\sqrt{N}$ gives the stated bound. $\square$

Corollary 10 (A nontrivial excess over the primes). *There is an absolute constant $c_0 > 0$ such that*

$$L(N) \geq \pi(N) + c_0 \frac{N^{3/4}}{(\log N)^{3/2}}$$

for infinitely many N , and in the usual asymptotic lower-bound sense

$$L(N) \geq \pi(N) + \Omega\left(\frac{N^{3/4}}{(\log N)^{3/2}}\right).$$

Proof. It is standard in extremal graph theory that there are m -vertex graphs with no C_3 and no C_4 and with $\Omega(m^{3/2})$ edges. Applying Proposition 9 with

$$m = \pi(\sqrt{N}) \sim \frac{2\sqrt{N}}{\log N}$$

gives

$$|E(G)| = \Omega(m^{3/2}) = \Omega\left(\frac{N^{3/4}}{(\log N)^{3/2}}\right).$$

The cost of removing the vertex primes is only

$$m = O\left(\frac{\sqrt{N}}{\log N}\right),$$

which is smaller than $m^{3/2}$. Hence the claimed lower bound follows. $\square$

Remark 11. The construction above is deliberately conservative. It uses only semiprimes with both prime factors at most $\sqrt{N}$. One can also use off-diagonal products $pq \leq N$, with one prime below and one prime above $\sqrt{N}$, using bipartite C_4 -free graphs between prime intervals. This can improve the lower-order constant, but it does not change the main unresolved issue: the exact second-order term appears to be governed by a weighted extremal graph problem.

7 A sensible follow-up problem

The proof of Theorem 2 resolves the main term but leaves a natural second-order problem. The semiprime construction suggests that the true excess over the primes should be on the same scale as in multiplicative Sidon theory:

$$\frac{N^{3/4}}{(\log N)^{3/2}}.$$

Problem 12 (Second-order LCM-Sidon problem). *Determine whether there is a constant $\kappa \geq 0$ such that*

$$L(N) = \pi(N) + (\kappa + o(1)) \frac{N^{3/4}}{(\log N)^{3/2}}.$$

If such a constant exists, determine it.

A more structural version is the following. Let Γ_N be the graph whose vertices are primes at most N , and where two distinct primes p, q are adjacent if

$$pq \leq N.$$

Define

$$M(N) = \max(|E(G)| - |V(G)|),$$

where the maximum is over all C_3, C_4 -free graphs $G \subseteq \Gamma_N$. The quantity $|E(G)| - |V(G)|$ is exactly the net gain obtained by deleting the vertex primes of G and inserting the semiprimes corresponding to its edges.

Problem 13 (Weighted girth-five reduction). *Is it true that*

$$L(N) = \pi(N) + M(N) + o\left(\frac{N^{3/4}}{(\log N)^{3/2}}\right)?$$

Equivalently, after the primes are accounted for, are all optimal lower-order gains essentially obtained by replacing selected primes with semiprimes from a C_3, C_4 -free graph on primes?

Remark 14. Problem 13 is stronger than merely asking for a better error term in Theorem 2. It asks for a structural theorem: every nearly extremal LCM-Sidon set should be reducible, after losing

$$o\left(\frac{N^{3/4}}{(\log N)^{3/2}}\right)$$

elements, to the prime set plus a semiprime graph construction. Such a result would convert the second-order number-theoretic problem into a weighted extremal graph problem.

Conjecture 15 (Expected qualitative second-order scale). *One has*

$$L(N) = \pi(N) + \Theta\left(\frac{N^{3/4}}{(\log N)^{3/2}}\right).$$

More ambitiously, Problems 12 and 13 should have affirmative answers with the same leading constant.

8 Summary

The original Problem 1 asked whether LCM-Sidon subsets of $[N]$ can have positive density. They cannot. The proof gives

$$L(N) = \pi(N) + O\left(N \exp(-c\sqrt{\log N})\right)$$

for an absolute $c > 0$, and hence

$$L(N) = (1 + o(1)) \frac{N}{\log N}.$$

The main mechanism is that each good element has a largest prime factor p , and the LCM-Sidon condition forces the corresponding fibres S_p to behave like a nearly disjoint family. This leaves room for at most one element per large prime factor, up to a small error.

The natural remaining direction is the second-order term. Semiprime constructions from C_3, C_4 -free graphs show that primes are not exactly optimal and give an excess of order at least

$$\frac{N^{3/4}}{(\log N)^{3/2}}.$$

The proposed follow-up is to prove that this is the correct scale and to identify the leading constant through a weighted extremal graph problem on primes.

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